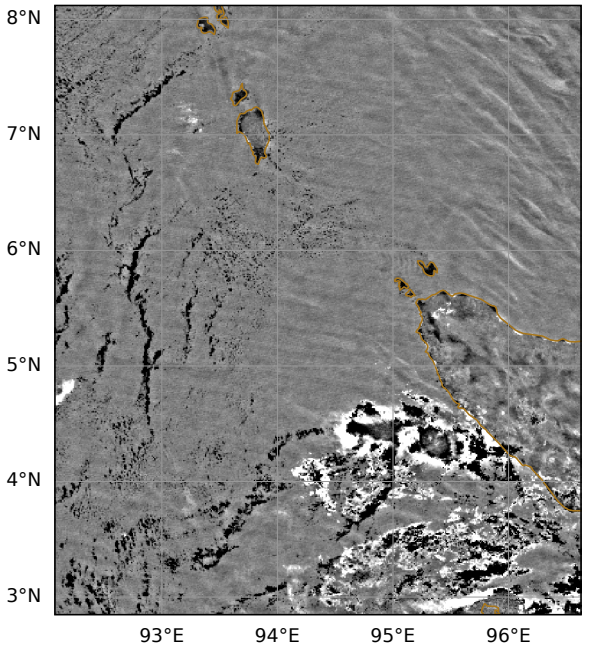
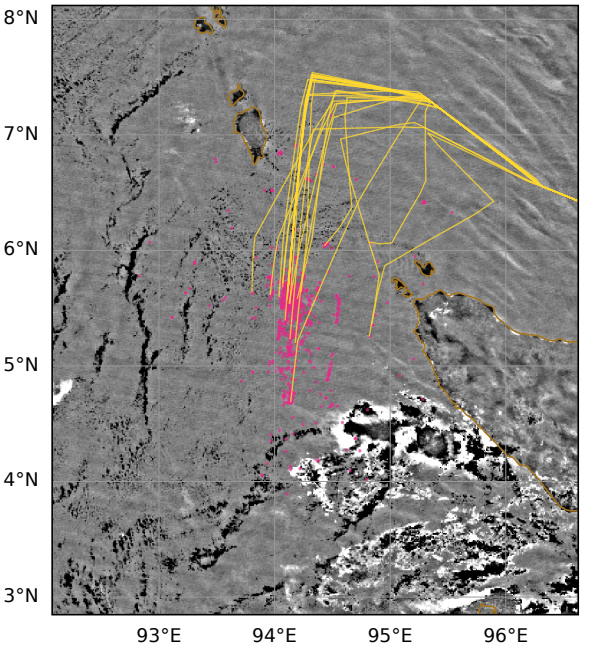
Suomi NPP VIIRS, 18:55 UTC 7 March 2014: no contrail visible in the split-window infrared image

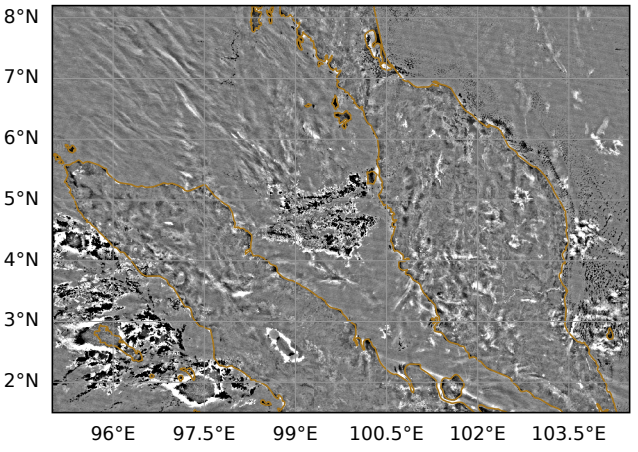
Likely aircraft area: image only



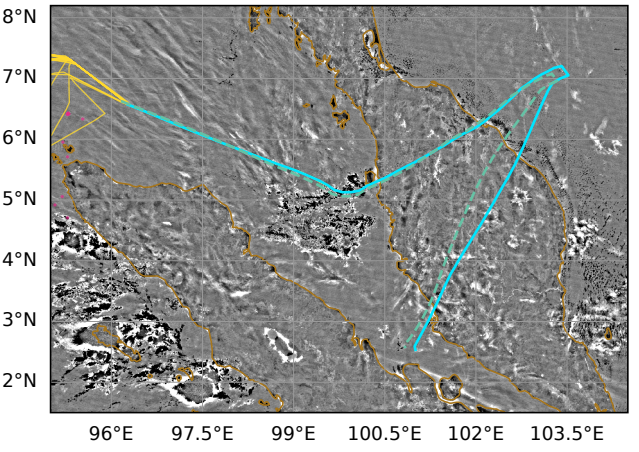
Likely aircraft area: with overlays

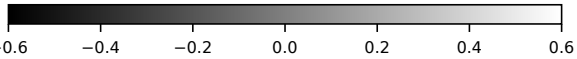


Radar-tracked flight area: image only



Radar-tracked flight area: with overlays





Local anomaly of the 11-12 um brightness temperature difference (K)

- likely aircraft position at the scan time (2,000 posterior samples)
- where a contrail up to 60 min old would lie (35 posterior routes, FL350 wind)
- where a contrail from the radar-tracked flight would lie, FL350 wind
- same, FL400 wind

Plain note: heat picture from Suomi NPP at 18:55 UTC. The pink dots are the likely aircraft positions. No contrail line is visible along the expected lines. The air was too warm for a contrail at FL350, so this is a weak test.

Technical note: VIIRS bands M15 (10.76 um) minus M16 (12.01 um), Collection 2 L1B (VNP02MOD.A2014066.1854.002, VNP03MOD.A2014066.1854.002; LAADS). Scan 18:55:31-18:55:39 UTC at the posterior; scan angle 32 deg; pixel 0.90 km along-track x 0.79 km along-scan; parallax 8.1 km; Sun -75 deg, Moon below the horizon (lunar zenith >= 109 deg in the scene, 42 % illuminated). Radar-tracked flight is inside the swath with contrail ages 33-133 min. Posterior: run J, BTO+BFO case (waypoints 0.9982, free 0.0013, reproduced-radar 0.0003, descent-climb 0.0002; 148,000 routes, linear interpolation between 600 s vertices); radar-tracked flight 16:42:27-18:22:12 from the Geoscience Australia digital military radar track. Positions use the granule's own scan time at each position and a parallax shift for FL350 (away from the sub-satellite point by 10.67 km x tan(view zenith)). Contrail loci use no contrail spreading; drift = ERA5 wind at FL350 (238.4 hPa) or FL400 (187.5 hPa) x contrail age. Image: brightness temperature difference minus its 21-pixel running median, on a 0.01 deg (VIIRS, MODIS) or 0.02 deg (AVHRR) nearest-neighbour grid. ERA5 (Schmidt-Appleman, efficiency 0.3): no contrail formation at FL300 or FL350 at this time; formation at FL400 but ice relative humidity < 100 %, so no persistence.